Course Code	Course Title	L	T	P	C
MACSE512	**Operating Systems**	**3**	**0**	**2**	**4**
Pre-requisite	Nil	**Syllabus Version**			
		1			

Course Objectives

1. To provide a comprehensive understanding of the evolution and fundamental components of modern operating systems, including process/thread management and virtualization technologies.
2. To equip students with the ability to design and analyze system-level mechanisms such as CPU scheduling, memory and file management, and synchronization in both traditional and real-time environments.
3. To prepare students for modern computing environments by exploring cloud OS concepts, containerization, and security techniques in distributed and edge systems.

Course Outcomes

1. Explain the structure and key components of an operating system including system calls, interfaces, and virtualization techniques.
2. Analyze process lifecycle, scheduling techniques, and implement synchronization mechanisms in multi-threaded environments.
3. Demonstrate memory management techniques such as segmentation, paging, and virtual memory across architectures.
4. Evaluate storage management approaches, disk scheduling algorithms, and file system implementation strategies.
5. Apply protection and security mechanisms in OS and explain the role of containers and distributed system management.

Module:1	**Introduction**	**5 hours**

Operating System Components – Design Architectures: monolithic, Layered, Microkernel, Modular, Hybrid – Operating System Operations: Interrupt Handling, Dual Mode Operation, Multitasking, Context Switching - Process Management -Memory Management - Storage Management - Protection and Security - Kernel Data Structures – Open Source Operating Systems: Definition, Types, Structures, Uses - Operating System Interface Types: Command Line, Graphical User, Touch Based - System Calls -Types of System Calls - System Programs - Virtualization Basics: OS-Level vs Hardware Level Virtualization – Hypervisors

Module:2	**Process Management**	**11 hours**

Process Concept - Process Scheduling - Operations on Processes - Inter-process Communication - Multicore Programming - Multithreading Models -Thread Libraries - CPU Scheduling - Scheduling Criteria - Scheduling Algorithms - Thread Scheduling -Multiple-Processor Scheduling - Real-Time OS Scheduling (Rate Monotonic, EDF) and Schedulability analysis - Process Synchronization - The Critical-Section Problem - Peterson's Solution - Synchronization Hardware -

Mutex Locks – Semaphores – Monitors -Classic Problems of Synchronization – Deadlock – Prevention – Avoidance- Detection – Recovery		
Module:3	**Memory Management**	**9 hours**
Contiguous Memory Allocation – Segmentation - Paging - Structure of the Page Table - Intel 32 and 64-bit Architectures - ARM Architecture - Virtual Memory - Demand Paging -Copy-on-Write - Page Replacement- Allocation of Frames - Thrashing - Allocating Kernel Memory		
Module:4	**Storage Management**	**9 hours**
Mass-Storage Structure - Disk Structure - Disk Attachment - Disk Scheduling - Disk Management - Swap-Space Management - File-System Interface - File Concept - Access Methods - Directory and Disk Structure - File-System Mounting - File Sharing - Protection - File-System Implementation -File-System Structure - Implementation - Directory - Allocation methods - NFS , EXT4 File Systems - I/O Systems - I/O Hardware - Application I/O Interface - Kernel I/O Subsystem - Distributed Storage Management		
Module:5	**Protection Security and Containerization**	**9 hours**
Protection - Domain of Protection - Access Matrix - Implementation of the Access Matrix -Access Control - Revocation of Access - Security - Program Threats- System and Network Threats - User Authentication - Implementing Security Defenses - Firewalling - Systems and Networks - Linux System Case study - Distributed Systems Basics - Containerization: Docker and Kubernetes		
Module:6	**Contemporary Issues**	**2 hours**
Contemporary Issues		
	Total Lecture Hours:	**45 hours**
Text Book(s)		
Abraham Silberschatz, Peter B. Galvin, Greg Gagne, " **“Operating System Concepts”"**, Wiley United States, 10th Edition, **2018** Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau , **"“Operating Systems: Three Easy Pieces”"**, Arpaci Dusseau Books, 1st Edition, **2023**		
Reference Books		
Thomas Anderson, Michael Dahlin, " **“Operating Systems: Principles and Practice”** ", Recursive Books Austin Texas USA, 2nd Edition, **2020** Kaiwan N. Billimoria , **"“Linux Kernel Programming”"**, Packt Publishing Birmingham United Kingdom, 2nd Edition, **2022**		
Indicative Experiments		

1. To familiarize students with basic Linux/Unix shell commands used for file handling, process management, user administration, and system navigation.	3 hours
2. To set up and configure a virtualization environment using tools such as Virtual Box, VMware, or KVM, and understand the basics of virtual machines and hypervisors.	3 hours
3. To implement and observe process creation and control using system calls such as fork(), exec(), wait(), and exit() in a Linux environment.	3 hours
4. To demonstrate the creation and execution of threads using the POSIX pthread library, and understand the basic concepts of multithreading and thread management.	3 hours
5. To simulate and compare CPU scheduling algorithms such as FCFS, SJF, Round Robin, and Priority Scheduling and analyze their performance based on turnaround time and waiting time.	3 hours
6. To simulate real-time scheduling algorithms such as Rate Monotonic Scheduling (RMS) and Earliest Deadline First (EDF), and evaluate their schedulability in real-time systems.	3 hours
7. To simulate deadlock detection in a multi-process system using resource allocation matrices and implement a deadlock detection algorithm (e.g., Banker's Algorithm or Wait-for Graph).	3 hours
8. To implement process synchronization using semaphores in producer-consumer or reader-writer problem scenarios and understand race conditions and critical section handling.	3 hours
9. To simulate dynamic memory allocation techniques such as First Fit, Best Fit, and Worst Fit and evaluate memory utilization and fragmentation.	3 hours
10. To simulate and evaluate page replacement algorithms such as FIFO, LRU, and Optimal and compare their performance in terms of page faults.	3 hours
Total Laboratory Hours:	**30 hours**
Mode of Evaluation : Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test, Lab Continuous Assessment, Lab Final Assessment	
Recommended by Board of Studies :	23-05-2025
Approved by Academic Council : No. 78	12-06-2025